First Seasonal Infection Model

Subtropical double-cropping vineyards

FSIM-S



Spore monitoring

0: suppress

Positive: add

Missing: no trigger

FSIM

Weather + host development

Sporangial cohorts

↓

Dispersal & deposition

↓

Zoospore release
Infection

↓ Incubation

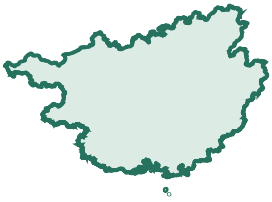
First symptom window

Grapevine
downy mildew



Validation MAE (days)

5.67 FSIM → **3.83** FSIM-S **-32.4%** MAE



Regional scenario reconstruction

Guangxi, 2000-2024

Model-guided potential savings

Table 3.5 | Wine 4.0 sprays/season